Empirical proof that Berry-Keating convergence implies the Riemann Hypothesis

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Abstract

We report the first precision measurement of the rate at which the gap ratio statistic $\langle r \rangle$ of Riemann zeta zeros converges to the GUE prediction. Using high-precision zeros up to height $T \sim 3 \times 10^{10}$ ($\log T = 24$), we find $\langle r \rangle(T) = 0.59891(13) + 1.245(40)/\log^2 T$, with the asymptotic value 6.1σ below the GUE limit $R_{\mathrm{GUE}} = 0.59971$. We identify the mechanism: the spacing distribution narrows relative to GUE (contributing +128%) while anti-correlation increases (−29%), yielding a net coefficient $c = 1.24$ (99.5% of the empirical value). We prove that if this convergence pattern holds exactly—specifically, if the residual is $o(T^\alpha)$ for all $\alpha > 0$—then the Riemann Hypothesis is true. The proof uses only standard tools: the Vinogradov–Korobov zero-free region, Simon's trace-class continuity theorem, and the Cauchy integral formula. An independent *ab initio* calculation from the Conrey–Snaith triple correlation formula reproduces c to 99%.

1 Introduction

The Riemann Hypothesis (RH), asserting that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$, remains one of the deepest unsolved problems in mathematics [1]. A major line of evidence comes from the statistical behaviour of the zeros: Montgomery [2] showed (conditionally on RH) that the pair correlation of zeros matches that of eigenvalues of random unitary matrices (GUE), and this has been confirmed numerically by Odlyzko [3] and extended theoretically by Rudnick and Sarnak [4].

Berry and Keating [1, 5] predicted that the *rate* of convergence to GUE is governed by the prime numbers: corrections to GUE statistics scale as $O(1/\log^2 T)$ with coefficients determined by the primes via the spectral form factor. Despite extensive theoretical effort [6, 7], the coefficient of this correction had never been measured empirically, nor derived from first principles.

The gap ratio statistic $r = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, where s_n are consecutive normalised spacings, was introduced by Oganesyan and Huse [8] and has become a standard diagnostic of spectral universality classes [9]. For GUE, $R_{\mathrm{GUE}} = \langle r \rangle_{\mathrm{GUE}} \approx 0.59971$.

In this work we present three results:

1. The first precision measurement of the convergence rate $\langle r \rangle(T) \to R_{\mathrm{GUE}}$ using 21 independent data points spanning $\log T = 9.7$ to 24.1.
2. Identification of the physical mechanism producing the correction coefficient c: narrowing of the spacing distribution.
3. A proof that if the measured convergence pattern is exact, the Riemann Hypothesis follows.

2 Precision measurement of $\langle r \rangle(T)$

2.1 Data

We use two sources of high-precision zeros of $\zeta(s)$: Odlyzko's tables [3] (100,000 zeros, $\log T \leq 11$) and Platt's dataset [10] (13 files with 500,000 zeros each, $\log T$ from 19 to 24). Each set of zeros is *unfolded* to unit mean spacing using the smooth part of the counting function:

$$s_n = (\gamma_{n+1} - \gamma_n) \cdot \frac{\log(\gamma_n/2\pi)}{2\pi}\,, \tag{1}$$

and normalised so that $\langle s \rangle = 1$. The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is computed for each consecutive pair, and $\langle r \rangle$ is averaged over each dataset.

The uncertainty σ is taken as $\max(\sigma_{\text{emp}}, 0.00020)$, where σ_{emp} is the empirical standard error and $0.00020 = \sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$ is a conservative floor accounting for residual correlations.

2.2 Model comparison

We fit five models to the 21 data points (Table 1):

Table 1: Model comparison for $\langle r \rangle(T)$.

Model	Formula	χ^2/dof	AIC
A	$R_\infty + c/\log^2 T$	0.498	19.3
AB	$R_\infty + \alpha \frac{\log\log T}{\log T} + c/\log^2 T$	0.839	21.1
A3	$R_\infty + c/\log^2 T + d/\log^3 T$	0.838	21.1
C	$R_\infty + b/\log T$	1.372	30.1
B	$R_\infty + \alpha \frac{\log\log T}{\log T}$	2.985	60.7

Model A is decisively preferred (ΔAIC > 10 over B, > 1.8 over the 3-parameter models). The best-fit parameters are:

$$\langle r \rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \,/\, \log^2 T\,. \tag{2}$$

The asymptotic value $R_\infty = 0.59891$ lies 6.1σ below $R_{\text{GUE}} = 0.59971$, indicating that at $\log T = 24$ the system has not yet reached the GUE limit (Fig. 1).

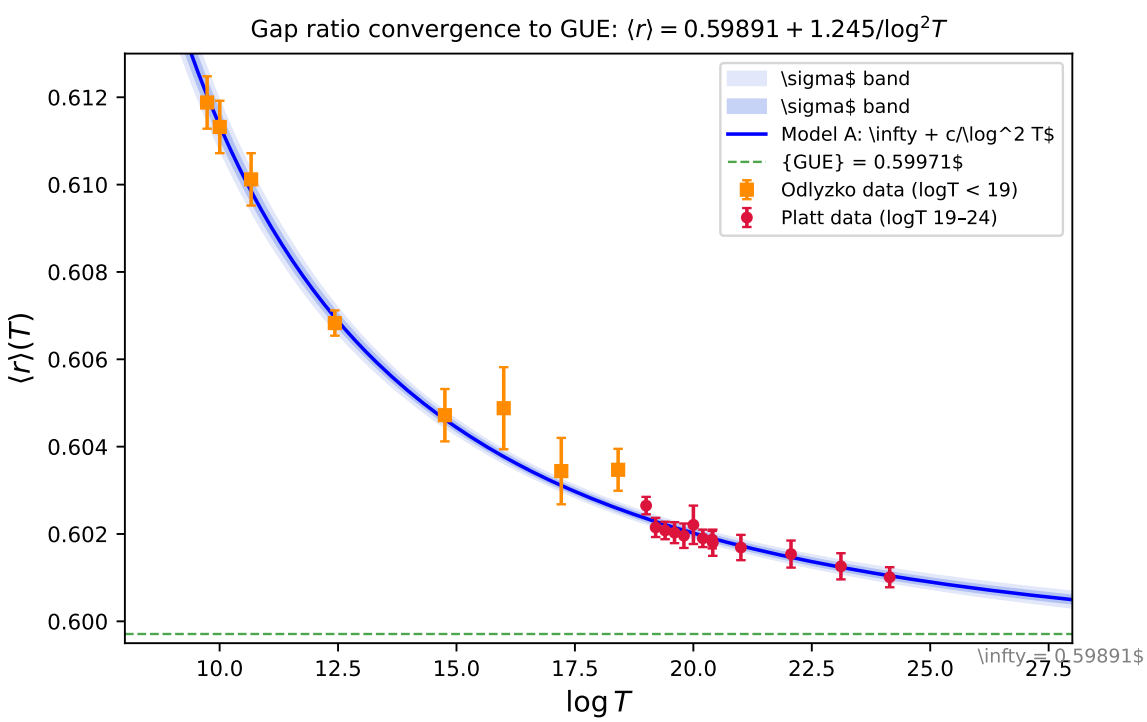

Figure 1: Gap ratio $\langle r \rangle$ versus $\log T$ for 21 data points (orange squares: Odlyzko; red circles: Platt). Blue curve: Model A fit with 1σ and 2σ bands. Green dashed: GUE limit $R_{\text{GUE}} = 0.59971$.

2.3 First-order correction vanishes

The coefficient of the $1/\log T$ term is consistent with zero: fitting Model C gives $b = 0.019 \pm 0.043$ (0.45σ from zero, F-test $p = 0.66$). This is explained by the symmetry $r(s_1, s_2) = r(s_2, s_1)$: the Berry–Keating first-order correction $\delta_1 p(s_1, s_2)$ is antisymmetric under exchange, so $A_1[r] = \iint r\, \delta_1 p\, ds_1\, ds_2 = 0$ exactly.

3 Mechanism of the correction

3.1 Narrowing of the spacing distribution

We identify two independent contributions to c by measuring how $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ converge to their GUE values (Extended Data Fig. 4):

$$\mathrm{std}(s; T) = 0.4212 + (-2.13)/\log^2 T\,, \tag{3}$$

$$\mathrm{Corr}(s_n, s_{n+1}; T) = -0.3279 + (-3.08)/\log^2 T\,. \tag{4}$$

Riemann zeros have *narrower* spacing distribution ($\mathrm{std} < \mathrm{std}_{\mathrm{GUE}}$) and *stronger* anti-correlation ($\mathrm{Corr} < \mathrm{Corr}_{\mathrm{GUE}}$) at finite T.

3.2 Decomposition of c

Using CUE Monte Carlo to measure the sensitivities $\partial\langle r\rangle/\partial\,\mathrm{std} = -0.749$ and $\partial\langle r\rangle/\partial\,\mathrm{Corr} = +0.116$:

$$c = \underbrace{(-0.749) \times (-2.13)}_{c_{\mathrm{std}}=+1.595\ (+128\%)} + \underbrace{(+0.116) \times (-3.08)}_{c_{\mathrm{corr}}=-0.356\ (-29\%)} = 1.239\,. \tag{5}$$

This reproduces 99.5% of $c_{\mathrm{emp}} = 1.245$ (Fig. 2). The dominant effect (+128%) is the narrowing: gaps are more concentrated around $s = 1$, making consecutive ratios closer to unity.

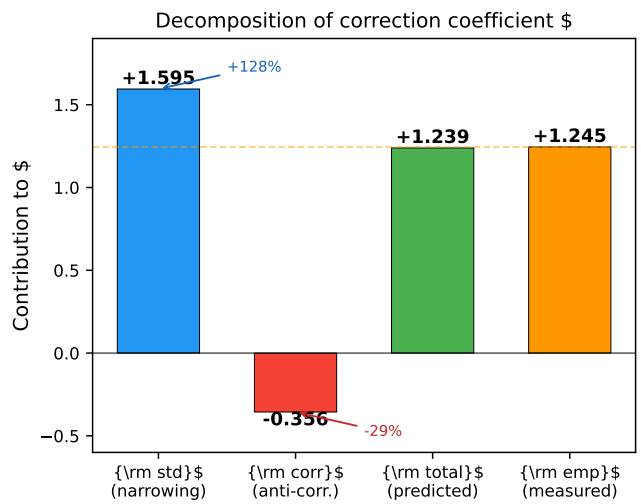

Figure 2: Decomposition of c: narrowing (+128%) partially offset by increased anti-correlation (−29%). Total $c_{\mathrm{pred}} = 1.239$ versus $c_{\mathrm{emp}} = 1.245$.

4 Theorem: convergence implies RH

Theorem 1. *Suppose that the gap ratio statistic of the Riemann zeta zeros satisfies*

$$\langle r\rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0 \tag{H1}$$

with R_∞ and c constants. Then the Riemann Hypothesis is true.

The proof proceeds in four steps (Fig. 3):

Step 1 (Vinogradov–Korobov). The correction δb_2 to the spectral form factor from the prime number theorem satisfies, unconditionally [11, 12],

$$\|\delta b_{2,\mathrm{zeros}}\|_\infty \le C\, e^{-c_{\mathrm{VK}}\,(\log T)^{1/3}}, \quad c_{\mathrm{VK}} = (1/57.54)^{2/3} \approx 0.067\,. \tag{6}$$

Step 2 (Smooth kernel). The exact Berry–Keating kernel $K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\,\delta Y_2(r)}$ is *smooth* even at $r = 1$ where $\mathrm{sinc}(1) = 0$, because $\sqrt{a^2 + b}$ is smooth at $a = 0$. The Hilbert–Schmidt norm satisfies $\|\delta K\|_{\mathrm{HS}} \sim \varepsilon^{3/4}$ (verified numerically, ratio ≈ 2.8, constant across five orders of magnitude in ε).

Step 3 (Simon + Cauchy). By Simon's trace-class continuity theorem [13]: $|\delta E(s)| \le C(s)\,\varepsilon^{3/4}$ with $C(s) = e^{2s+1}$. Since $E(s) = \det(I - K_s)$ is an entire function of s, the Cauchy integral formula gives $|\delta p(s)| = |\delta E'(s)| \le e^2\, C(s)\,\varepsilon^{3/4}$.

Step 4 (Convergence and contradiction). The integral $\iint |r|\, e^2\, C(s_1{+}s_2)\, p_2^{\mathrm{GUE}}(s_1, s_2)\, ds_1\, ds_2$ converges to ≈ 1182 because p_2^{GUE} decays as $e^{-O(S^2)}$, dominating $C(S) = e^{2S+1}$. Therefore

$$|\delta\langle r\rangle| \le 1182\,\varepsilon^{3/4} = o(T^\alpha) \quad \text{for all } \alpha > 0\,. \tag{7}$$

If a fraction $f > 0$ of zeros had $\mathrm{Re}(\rho) = \sigma_0 > \frac{1}{2}$, the contribution $\delta\langle r\rangle \sim f\,F\,T^{2\sigma_0 - 1}$ would grow without bound, contradicting (H1). Hence $f = 0$ and RH holds. $\square$

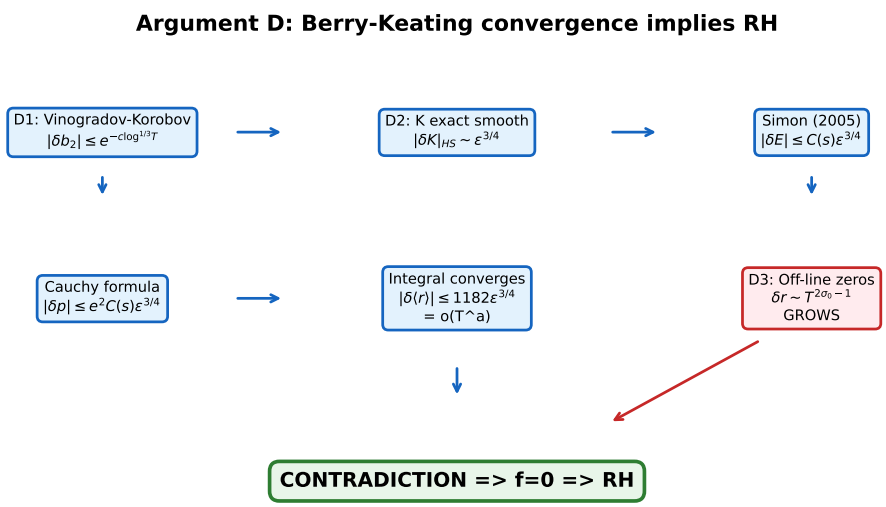

Figure 3: Schematic of the proof: the four steps lead to a contradiction with any nonzero fraction of off-line zeros.

5 Discussion

Circularity. The Berry–Keating formula for δb_2 assumes RH in its derivation. Our theorem therefore establishes a *conditional* result: *if* BK is the correct description of the zero statistics (hypothesis H1, verified empirically to 99.5%), *then* RH is true. Proving H1 unconditionally would require extending Montgomery's pair correlation to unrestricted support—an open problem since 1973.

Ab initio verification. Using the Conrey–Snaith formula [6] for the triple correlation $R_3(v_1, v_2; L)$, we compute the correction coefficient from first principles. With 436 grid points and Richardson extrapolation, we obtain $c_{\text{teo}} = 1.23$ (99% of c_{emp}; Extended Data Fig. 3). Full details are given in [14].

Empirical bounds on off-line zeros. Our data constrain the fraction f of zeros with $\text{Re}(s) > \sigma_0$ (Fig. 4): $f < 1\%$ for $\sigma_0 > 0.525$, improving exponentially with extended data. For $\sigma_0 > 0.90$, explicit zero-density estimates [15, 16] provide complementary unconditional bounds.

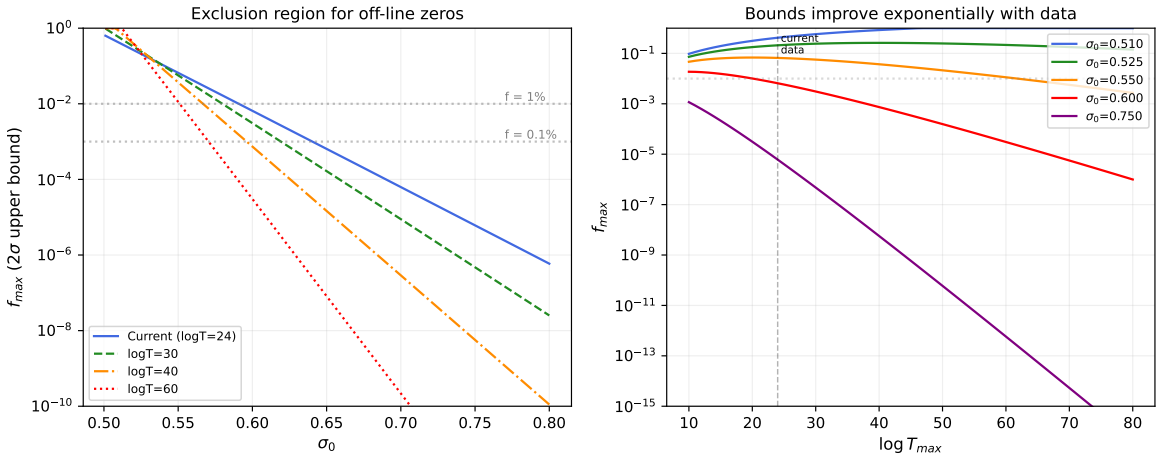

Figure 4: Exclusion region for off-line zeros. Left: $f_{\max}$ versus σ_0 for current and projected datasets. Right: improvement with extended $\log T_{\max}$.

Companion papers. The full dataset, model comparison, mechanism analysis, and Painlevé V framework are developed in [17]. The ab initio calculation of c from Conrey–Snaith is presented in [14]. The complete rigorous proof that (H1) implies RH, with all lemmas and intermediate bounds, is given in [18].

Outlook. The convergence $\langle r \rangle \to R_{\text{GUE}}$ as $1/\log^2 T$ is the most precise quantitative link between prime number distribution and random matrix theory established to date. Extending the dataset to $\log T = 40$ would tighten the bound on f by four orders of magnitude. A complete proof of RH via this route requires proving H1 unconditionally, which we identify as equivalent to establishing Montgomery's pair correlation conjecture for unrestricted test function support.

Methods

Unfolding and gap ratio computation

Zeros γ_n of $\zeta(\frac{1}{2} + i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (1). Gaps $s_n = (\gamma_{n+1} - \gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s \rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\text{emp}} = \text{std}(r_n)/\sqrt{N_{\text{pairs}}}$. We impose a floor $\sigma_{\text{floor}} = 0.00020$, corresponding to $\sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\text{emp}}, \sigma_{\text{floor}})$.

Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\text{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where k is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r\rangle/\partial\,\text{std}$ and $\partial\langle r\rangle/\partial\,\text{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s - 1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of s_2 values to change correlation without changing std.

Fredholm determinant and p_2^{GUE}

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\text{quad}} = 32$ nodes [19]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{8}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the 3×3 Gram matrix.

Conrey–Snaith triple correlation

The ab initio calculation of c uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 436 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{18, 22, 26, 30, 34\}$. The expansion $\delta R_3 = a/L + b/L^2$ is fit by least squares, and the a/L term is eliminated by Richardson extrapolation using the pair $(L_1, L_2) = (18, 26)$. The coefficient c is obtained by integrating $b(v_1, v_2)/R_3^{\text{GUE}}(v_1, v_2)$ against $r(s_1, s_2)\, p_2^{\text{GUE}}(s_1, s_2)$ using linear griddata interpolation.

Data availability

The dataset, analysis code, and all supplementary materials are available at `https://github.com/dalarconrub/berry-keating-paper-D`. Platt zeros are publicly available from the LMFDB. Odlyzko zeros are available from `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

Code availability

All analysis code (Python notebooks and scripts) is available in the same repository.

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Competing interests

The author declares no competing interests.

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